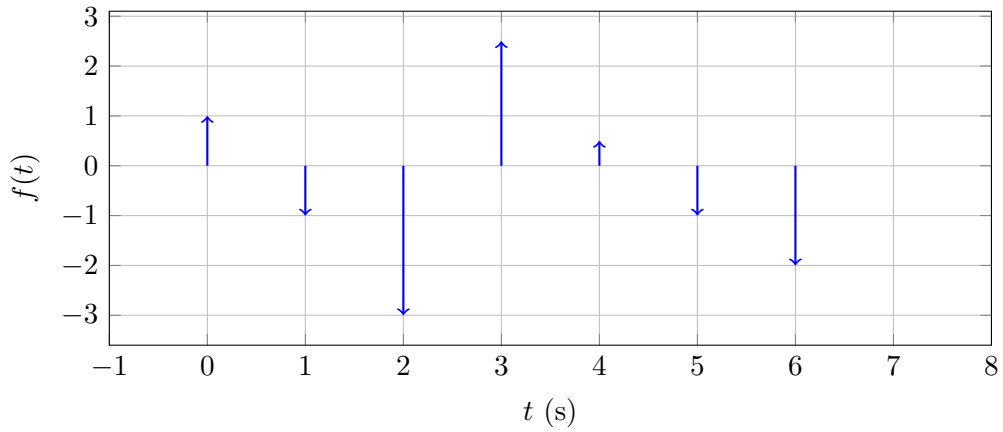
ECE 3210 hw06

1. A zero-order hold reconstructs a signal from the samples

$$\bar{f}(t) = \sum_{n=0}^{6} a_n \delta(t-n), \qquad (a_0, \ldots, a_6) = (1, -1, -3, 2.5, 0.5, -1, -2),$$

using the impulse response $h(t) = \text{rect}(t)$, where $\text{rect}(t) = 1$ for $|t| \leq \frac{1}{2}$ and is zero otherwise. $\bar{f}(t)$ is shown in the figure below with samples taken in 1 second intervals.



(a) Write the reconstructed signal $f(t)$.

Solution: The reconstructed signal is given by

$$f(t) = \sum_{n=0}^{6} a_n \, \text{rect}(t-n),$$

where $(a_0, \ldots, a_6) = (1, -1, -3, 2.5, 0.5, -1, -2)$.

(b) Determine the frequency response $H(\omega)$ of the hold.

Solution: This is simply the Fourier transform of the impulse response $h(t)$

$$H(\omega) = \text{sinc}(\omega/2)$$

based on the table.

(c) Find the minimum delay required to make the hold causal.

Solution: For the hold to be causal, the impulse response $h(t)$ must be zero for all $t < 0$. Since $h(t) = \text{rect}(t)$, we have $h(t) = 1$ for $|t| \leq \frac{1}{2}$ and is zero otherwise. Therefore, the minimum delay required is $T_d = \frac{1}{2}$ seconds.

(d) Give the delayed reconstructed signal $f_d(t)$.

Solution: The delayed reconstructed signal is given by

$$f_d(t) = f(t - \tfrac{1}{2}).$$

This means each rectangle is simply shifted by 1/2 second to the right, making the hold causal.

(e) Determine the frequency response $H_d(\omega)$ after the delay.

Solution: The frequency response after the delay is given by

$$H_d(\omega) = e^{-j\omega/2} H(\omega) = e^{-j\omega/2} \operatorname{sinc}(\omega/2).$$

2. Let $f_1(t)$ and $f_2(t)$ be bandlimited signals with bandwidths of 150 Hz and 75 Hz, respectively. Also consider a sinusoid sampled and reconstructed by an ideal low-pass filter, and a first-order-hold reconstruction system.

   (a) Find the Nyquist sampling rate for $f_1(t) - f_2(t)$.

   Solution: We see that the bandwidth of $f_1(t) - f_2(t)$ is the maximum of the two bandwidths, which is 150 Hz. Therefore, the Nyquist sampling rate is twice the bandwidth

   $$f_{\text{Nyq}} = 2 \times 150 \text{ Hz} = 300 \text{ Hz}.$$

   (b) Find the Nyquist sampling rate for $f_1^2(t)$.

   Solution: If signals are multiplied in time, then the Fourier transforms are convolved. The convolution width property kicks in and so the bandwidths simply add. Therefore, the bandwidth of $f_1^2(t)$ is $150 + 150 = 300$ Hz, and the Nyquist sampling rate is

   $$f_{\text{Nyq}} = 2 \times 300 \text{ Hz} = 600 \text{ Hz}.$$

   (c) A sinusoid $x(t) = A\sin(2\pi f_0 t)$ with $f_0 = 7700$ Hz is sampled at $f_s = 8000$ Hz and reconstructed with an ideal low-pass filter having cutoff $f_c = f_s/2$. What tone frequency is heard?

   Solution: The Nyquist frequency is $f_s/2 = 4000$ Hz. The original frequency $f_0 = 7700$ Hz is above the Nyquist frequency, so it will alias. The aliased frequency can be found using the formula

   $$f_{\text{alias}} = |f_0 - n f_s|,$$

   where n is an integer chosen such that f_{alias} falls within the range $[0, f_s/2]$.
   For $n = 1$, we have

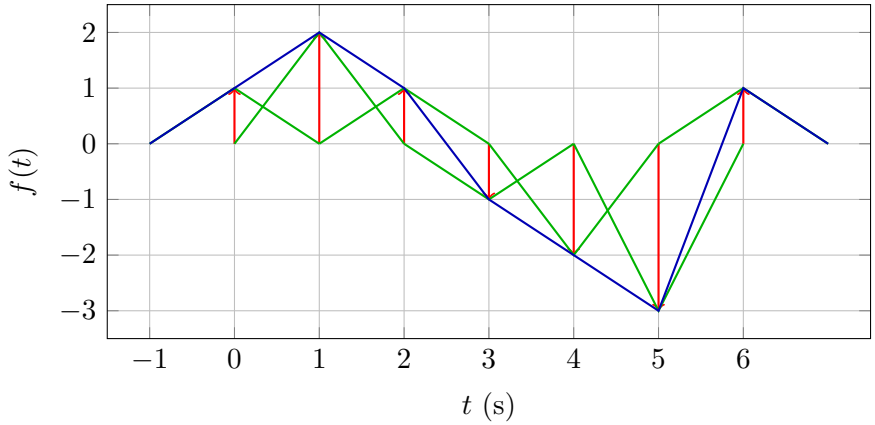
   $$f_{\text{alias}} = |7700 - 8000| = 300 \text{ Hz}.$$

   Therefore, the tone frequency heard after reconstruction is 300 Hz.

   (d) A first-order hold with sampling interval $T = 1$ s has impulse response $h(t) = \Delta(t/2)$, where $\Delta(u) = \max(1 - 2|u|, 0)$. The samples are $\bar{f}(t) = \sum_{n=-1}^{6} a_n \delta(t - n)$, with $(a_{-1}, \ldots, a_6) = (0, 1, 2, 1, -1, -2, -3, 1)$. Write the reconstructed signal $f(t)$ and sketch it along with the sample values.

   Solution: The sampled signal is seen in the red arrows. The green triangles show each of the shifted and scaled impulse responses. The blue step trace

shows the reconstructed signal $f(t)$, which is the sum of the green triangles.



(e) Determine the frequency response $H(\omega)$ of the first-order hold.

Solution: This is just the Fourier transform of the impulse response $h(t)$

$$H(\omega) = \text{sinc}^2(\omega/2).$$

(f) Find the minimum delay required to make the filter realizable, and give the resulting frequency response $H_d(\omega)$.

Solution: To make this causal, we need to delay the impulse response by $T_d = 1$ second. The delayed frequency response is then

$$H_d(\omega) = e^{-j\omega T_d} H(\omega) = e^{-j\omega} \text{sinc}^2(\omega/2).$$

(g) How does the delay affect the reconstructed signal?

Solution: The delay causes the reconstructed signal to be shifted in time by $T_d = 1$ second. This means that the reconstructed signal $f_d(t)$ is equal to the original signal $f(t)$ shifted by one second.

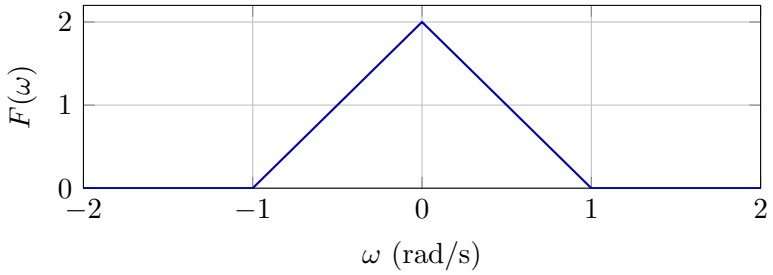
3. Consider the signal

$$f(t) = \frac{1}{\pi}\,\mathrm{sinc}^2\left(\frac{t}{2}\right).$$

(a) Sketch its Fourier transform $F(\omega)$.

Solution: We can solve this using the book table, which gives

$$F(\omega) = 2\Delta\left(\frac{\omega}{2}\right).$$

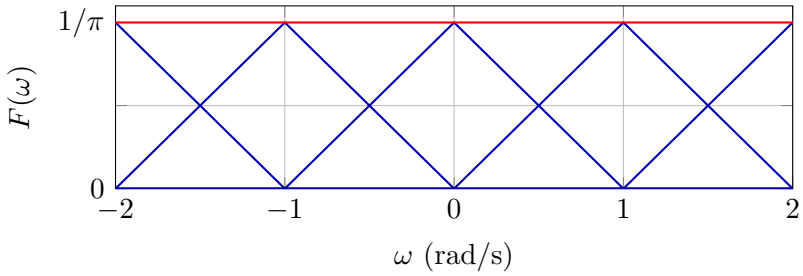


(b) Find the signal bandwidth in hertz and the lowest sampling rate that avoids aliasing.

Solution: We see the bandwidth is $B = 1/(2\pi)$ Hz, and the Nyquist sampling rate is $f_{\text{Nyq}} = 1/\pi$ Hz.

(c) Suppose the signal is sampled at one-half the Nyquist rate. Find the Fourier transform $\bar{F}(\omega)$ of the sampled signal.

Solution: If we sample at one-half the Nyquist rate, we have $f_s = 1/(2\pi)$ Hz. This would mean that we would see copies of the spectrum at $\pm n \cdot \omega_s$ where $\omega_s = 2\pi f_s = 1$ rad/s. We will therefore have copies at $\pm n$ where n is an integer. Additionally, we will need to scale by $1/T_s$ where $T_s = 1/f_s = 2\pi$ s. Therefore, the Fourier transform of the sampled signal is a sum of shifted copies of the original spectrum, scaled by $\frac{1}{T_s} = \frac{1}{2\pi}$. Thus, each copy of the spectrum will have a maximum value of $1/\pi$ and will overlap with adjacent copies, resulting in aliasing.



(d) Find an expression for $\bar{f}(t)$, the sampled signal in the time domain (using delta functions).

Solution: We see that we are sampling at $\omega_s = 1$ rad/s, corresponds to the zero crossings of the original $f(t)$ squared sinc function! Essentially, the sample at $n = 0$ is non-zero $(1/\pi)$, and all other samples are zero. Therefore, the sampled signal is simply a single delta function at $t = 0$ scaled by $1/\pi$.

(e) Reconstruct the sampled signal using an ideal low-pass filter with cutoff frequency one-quarter of the Nyquist sampling rate and unity passband gain. Find the resulting signal $f_{\text{recon}}(t)$.

Solution: If we are using $1/4$ of the Nyquist sampling rate, then we have $\omega_c = 1/2$ rad/s. The ideal low-pass filter in the frequency domain would be

$$\text{rect}\,(\omega)$$

The reconstruction in the frequency domain is simply the product of the sampled signal and the low-pass filter. The sampled signal in frequency is a constant $1/\pi$ for all ω, and the low-pass filter is 1 for $|\omega| < 1/2$ and 0 otherwise. Therefore, the product will be

$$F_{\text{recon}}(\omega) = \frac{1}{\pi}\,\text{rect}(\omega)$$

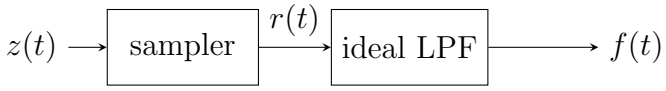
Taking in the inverse Fourier transform, we have

$$f_{\text{recon}}(t) = \frac{1}{2\pi^2}\,\text{sinc}(t/2).$$

4. Consider the message signal

$$f(t) = \frac{1}{2\pi}\,\mathrm{sinc}^2\left(\frac{t}{4}\right)$$

which is modulated by a carrier according to $z(t) = f(t)\cos(2t)$. The receiver samples $z(t)$ and then applies an ideal low-pass filter to recover $f(t)$.



(a) Determine the Fourier transform $F(\omega)$.

Solution: Using the table, we see that the Fourier transform of $f(t)$ is

$$F(\omega) = 2\Delta(\omega).$$

(b) Determine the Fourier transform $Z(\omega)$ of the modulated signal.

Solution: See that $z(t) = f(t)\cos(2t)$, so we can use the modulation property of the Fourier transform to find $Z(\omega)$

$$Z(\omega) = \frac{1}{2}\left[F(\omega - 2) + F(\omega + 2)\right].$$

and then

$$Z(\omega) = \Delta(\omega - 2) + \Delta(\omega + 2).$$

(c) For a sampling rate $f_s = 1/\pi$ Hz, find the sampling period T_s, sampling angular frequency ω_s, and the spectrum $R(\omega)$ of the sampled signal.

Solution: If the sampling rate is $f_s = 1/\pi$ Hz, then the sampling period is

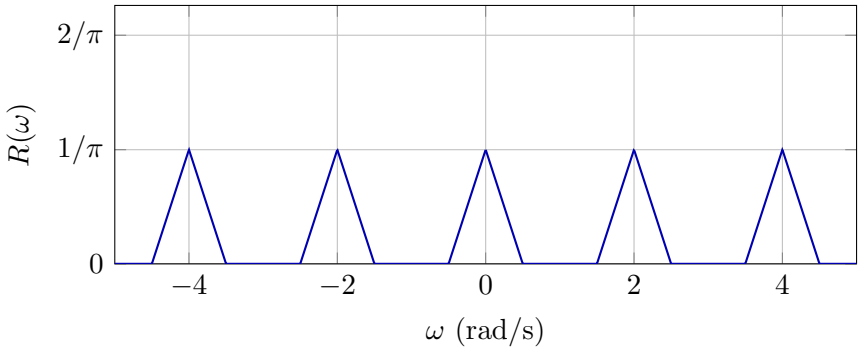
$$T_s = \frac{1}{f_s} = \pi \text{ s}.$$

The sampling angular frequency is

$$\omega_s = 2\pi f_s = 2.$$

The spectrum of the sampled signal is given by

$$R(\omega) = \frac{1}{T_s}\sum_{k\in\mathbb{Z}} Z(\omega - k\omega_s) = \frac{1}{\pi}\sum_{k\in\mathbb{Z}} Z(\omega - 2k).$$



(d) What is the largest sampling period T that still permits recovery of $f(t)$ using this receiver?

Solution: We see that the shortest sampling rate is $\omega_s = 1$ won't lead to aliasing. Thus, the largest sampling period is

$$T_{\max} = \frac{2\pi}{\omega_s} = \frac{2\pi}{1} = 2\pi \text{ s.}$$

5. Consider the bandlimited signal

$$f(t) = \frac{1}{\pi}\operatorname{sinc}^2\left(\frac{t}{2}\right)$$

(a) Determine the Fourier transform $F(\omega)$.

Solution: Same process as the previous problem. Let's use the tables and we see that

$$F(\omega) = 2\Delta\left(\frac{\omega}{2}\right).$$

(b) Find the bandwidth in hertz and the Nyquist sampling rate.

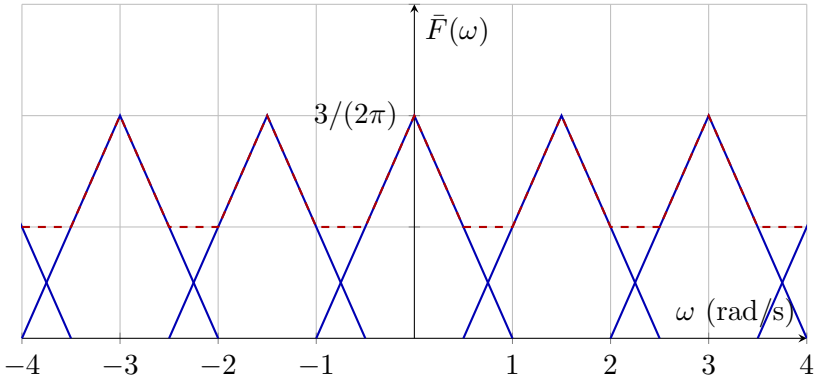
Solution: The baseband bandwidth is $B = \frac{1}{2\pi}$ Hz and the Nyquist sampling rate is $f_{\text{Nyq}} = \frac{1}{\pi}$ Hz.

(c) The signal is sampled at three-quarters of the Nyquist sampling rate. Plot the Fourier transform of the sampled signal $\bar{F}(\omega)$.

Solution: The Nyquist sampling rate in radians per second is $\omega_{\text{nyq}} = 2$ rad/s. If we use a sampling rate that is three-quarters of the Nyquist rate, then $\omega_s = 1.5$ rad/s. The sampling period is $T = \frac{1}{f_s} = \frac{2\pi}{\omega_s} = \frac{2\pi}{1.5} = \frac{4\pi}{3}$ s. The Fourier transform of the sampled signal is given by

$$\bar{F}(\omega) = \frac{1}{T}\sum_{k=-\infty}^{\infty} F(\omega - k\omega_s) = \frac{1.5}{2\pi}\sum_{k=-\infty}^{\infty} F(\omega - k(1.5)).$$

We plot this below. The dashed red line is the final answer.



(d) Instead, modulate the original signal by $\cos(0.5t)$, so that $f_m(t) = f(t)\cos(0.5t)$. Determine $F_m(\omega)$.

Solution: We can use the modulation property of the Fourier transform. If $f(t)$ has a Fourier transform $F(\omega)$, then $f(t)\cos(\omega_0 t)$ has a Fourier transform

$$F_m(\omega) = \frac{1}{2}\left[F(\omega - \omega_0) + F(\omega + \omega_0)\right].$$

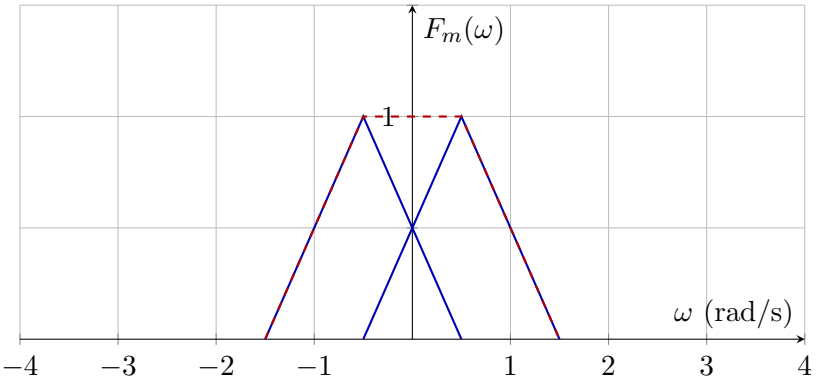
In this case, $\omega_0 = 0.5$, so we have

$$F_m(\omega) = \frac{1}{2}\left[F(\omega - 0.5) + F(\omega + 0.5)\right].$$

Using $F(\omega) = 2\Delta\left(\frac{\omega}{2}\right)$, we have

$$F_m(\omega) = \frac{1}{2}\left[2\Delta\left(\frac{\omega - 0.5}{2}\right) + 2\Delta\left(\frac{\omega + 0.5}{2}\right)\right] = \Delta\left(\frac{\omega - 0.5}{2}\right) + \Delta\left(\frac{\omega + 0.5}{2}\right).$$

We should probably plot this so it makes sense. The plot is below. The dashed red line is the final answer.



6. Consider the complex-valued signal

$$f(t) = \frac{1}{2}\,\mathrm{sinc}^2\left(\frac{\pi t}{2}\right) e^{j\pi t}$$

(a) Derive the Fourier transform $F(\omega)$.

Solution: First, let's look at signal $\tilde{f}(t) = \frac{1}{2}\,\mathrm{sinc}^2\left(\frac{\pi t}{2}\right)$. The Fourier transform of $\tilde{f}(t)$ is given by

$$\tilde{F}(\omega) = \Delta\left(\frac{\omega}{2\pi}\right).$$

If we modulate in time by multiplying by $e^{j\pi t}$, we shift the spectrum in frequency by π. Therefore, the Fourier transform of $f(t)$ is

$$F(\omega) = \tilde{F}(\omega - \pi) = \Delta\left(\frac{\omega - \pi}{2\pi}\right).$$

Importantly, because $f(t)$ is complex-valued, we lose Hermitian symmetry in the spectrum, so the magnitude of $F(\omega)$ will not be even symmetric.

(b) Using the highest frequency represented in the signal as its bandwidth, find the bandwidth B in hertz.

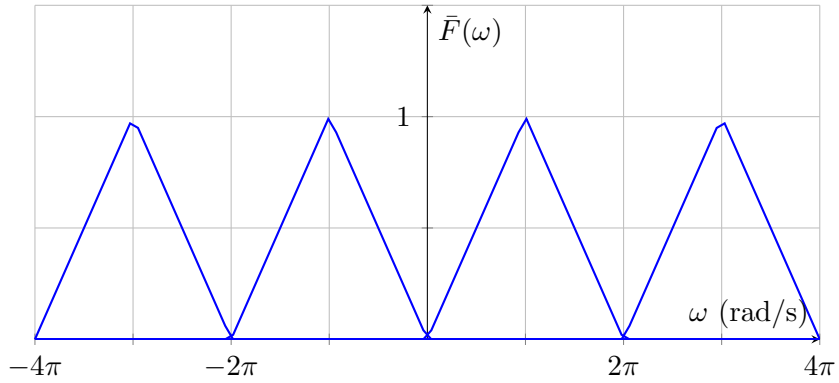
Solution: The bandwidth B is defined as the highest frequency component present in the signal. From the Fourier transform $F(\omega)$, we see that the non-zero frequency components are centered around $\omega = \pi$ and extend from $\omega = 0$ to $\omega = 2\pi$. Therefore, the highest frequency represented in the signal is $\omega = 2\pi$ radians per second. To convert this to hertz, we get $B = \frac{2\pi}{2\pi} = 1$ Hz.

(c) Find the longest sampling period $T_{\max}$ that avoids aliasing and the corresponding sampling rate f_s.

Solution: It would be helpful to sketch this as we determine this value, but the answer here is $\omega_s = 2\pi$ rad/s, which corresponds to $f_s = 1$ Hz and $T_{\max} = 1$ s.

(d) Sketch the spectrum $\bar{F}(\omega)$ of the sampled signal and state whether the original signal can be recovered from these samples.

Solution: First we see that the original spectrum can be thought of as spanning from -2π to 2π in ω, though it is zero from -2π to 0. The sampling frequency is $\omega_s = 2\pi$, so the replicas of the spectrum will be centered at integer multiples of 2π.



From the sketch, we can see that the replicas of the spectrum do not overlap, which means that the original signal can be recovered from these samples. We would need to be clever about how to recover the original signal, since it is complex-valued and not Hermitian symmetric, but it is possible.